Reading time: 15 minutes — Writing time/Duration: 50 minutes
(the real exam is 120 minutes)

This practice exam consists of only 6 questions and 50 marks
(the real exam has more questions and it is 120 marks).

Permitted Materials

- Identification card (Hardcopy student ID, Passport or Australian Government-issued photo ID)
- Pencils, pens, erasers
- Clear or unlabelled bottle of water

Section A: Short Answer Questions [10 marks]

Answer each of the questions in this section as briefly as possible. Expect to answer each question in 1-3 lines, with longer responses expected for the questions with higher marks.

Question 1: [10 marks]

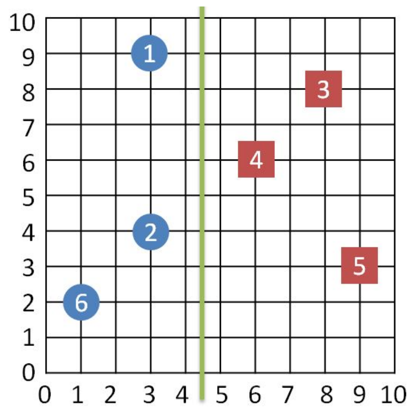
(a) In words or a mathematical expression, what quantity is minimised by *linear regression*? **[2 marks]**

Write your answers below:

(b) In words or a mathematical expression, what is the *marginal likelihood* for a *Bayesian probabilistic model*? **[2 marks]**

Write your answers below:

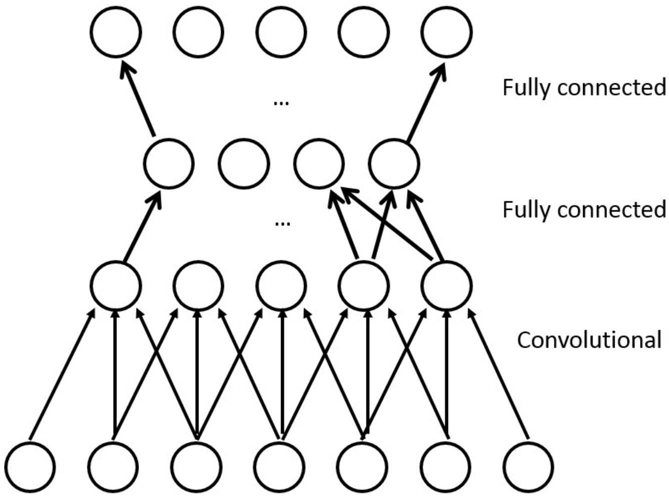
(c) Consider the data shown below with *hard-margin linear SVM decision boundary* shown between the classes. The right half is classified as red squares and the left half is classified as blue circles.



Which points (by index 1–6) would be the *support vectors* of the *SVM*? **[3 marks]**

Write your answers below:

(d) How many *parameters* does the following *convolutional neural network* have (exclude the *bias*)? **[3 marks]**



Write your answers below:

Section B: Method & Calculation Questions [30 marks]

In this section you are asked to demonstrate your conceptual understanding of methods that we have studied in this subject, and your ability to perform numeric and mathematical calculations.

Question 2: [6 marks]

You are running a multi-armed bandit problem with 3 arms. The rewards from each arm are binary (0 or 1), and so far, you have pulled the arms with the following results:

- Arm 1: Pulled 64 times, with 40 rewards of 1.
- Arm 2: Pulled 16 times, with 10 rewards of 1.
- Arm 3: Pulled 4 time, with 2 reward of 1.

Using the UCB algorithm, calculate the value of each arm based on the given results. You can provide your answer in fractions or round your answers to 3 decimal places. Based on your calculations, which arm should you pull next, and why?
Some possibly useful values: $\ln(4) \approx 1.4, \ln(5) \approx 1.6, \ln(16) \approx 2.8, \ln(17) \approx 2.8, \ln(64) \approx 4.2, \ln(65) \approx 4.2, \ln(84) \approx 4.5, \ln(85) \approx 4.5$
Some possibly useful formulas:

$$Q_{t-1}(i) = \widehat{\mu}_{t-1}(i) + \sqrt{\frac{2\ln t}{N_{t-1}(i)}}, \quad \text{where } N_{t-1}(i) = \sum_{s=1}^{t-1} 1[i_s = i], \quad \widehat{\mu}_{t-1}(i) = \frac{\sum_{s=1}^{t-1} R_i(s) 1[i_s = i]}{N_{t-1}(i)}.$$

Write your answers below:

Question 3: [8 marks]

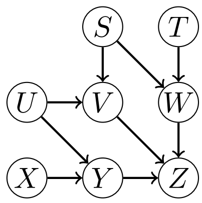
Consider an *input domain* of five points $x_1, \ldots, x_5$, and *binary classifier family* $\mathcal{G}$ defined by the table of *dichotomies* given below. Calculate $\mathrm{VC}(\mathcal{G})$ and show a corresponding shattered set of points.

x_1	x_2	x_3	x_4	x_5
0	0	0	0	1
1	0	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	0	1	1
1	0	1	1	1
1	1	1	0	1
0	1	1	1	1

Write your answers below:

Question 4: [8 marks]

Consider the following Bayesian network:

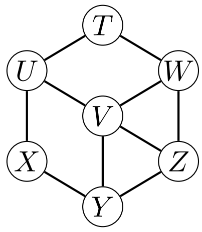


Does the conditional independence statement $S \perp Z \mid V, W$ hold in this Bayesian network?

Write your answers below:

Question 5: [8 marks]

Consider the following Markov Random field:



Assume the following node elimination order: T, X, Z, U, W, V, Y. (Eliminate the first node in the list, then eliminate the second node in the list, then eliminate the third node in the list, etc.) What are the cliques in the junction tree (by using the above elimination order)?

Write your answers below:

Section C: Design and Application Questions [10 marks]

In this section you are asked to demonstrate that you have gained a high-level understanding of the methods and algorithms covered in this subject, and can apply that understanding.

Question 6: [10 marks]

Your task is to design a system for alerting residents of the Dandenongs that they should evacuate for an impending bushfire. The Dandenongs is an area of Victoria that suffers from regular bushfires in the summer when temperatures are high, and humidity low. When fires are close to a fictional town called Bayesville, you must alert residents that they should evacuate. If fires are too close, then you should not advise evacuation as residents are safer if they stay where they are (at home).
The Country Fire Association has deployed sensors around the area that monitor whether a fire is in progress at each sensor's location; in particular if any of three sensors S_1, S_2, S_3 are 'on' then residents should evacuate. However if either of the closer sensors C_1, C_2 are 'on' then residents should stay put.

(a) Model the above problem as a *directed probabilistic graphical model* (PGM). In particular, you need not provide any probabilty tables, just the graph relating random variables S_1, S_2, S_3, C_1, C_2 and an additional r.v. A for alerting residents to evacuate. **[4 marks]**

Write your answers below:

(b) How many *conditional probability tables (CPTs)* should be specified for your model, and what should these tables' dimensions be? **[3 marks]**

Write your answers below:

(c) Suppose you are told by the Fire Commissioner that the sensors are always accurate. What could you say about your CPTs? **[3 marks]**

Write your answers below:

— *End of Exam* —